

** In order for this comparison to succeed, it is important to account for the inclination angle of the star-knot separation vector with respect to the plane of the sky. High values of the inclination mean that the true separation is larger than the projected separation (Fig I).

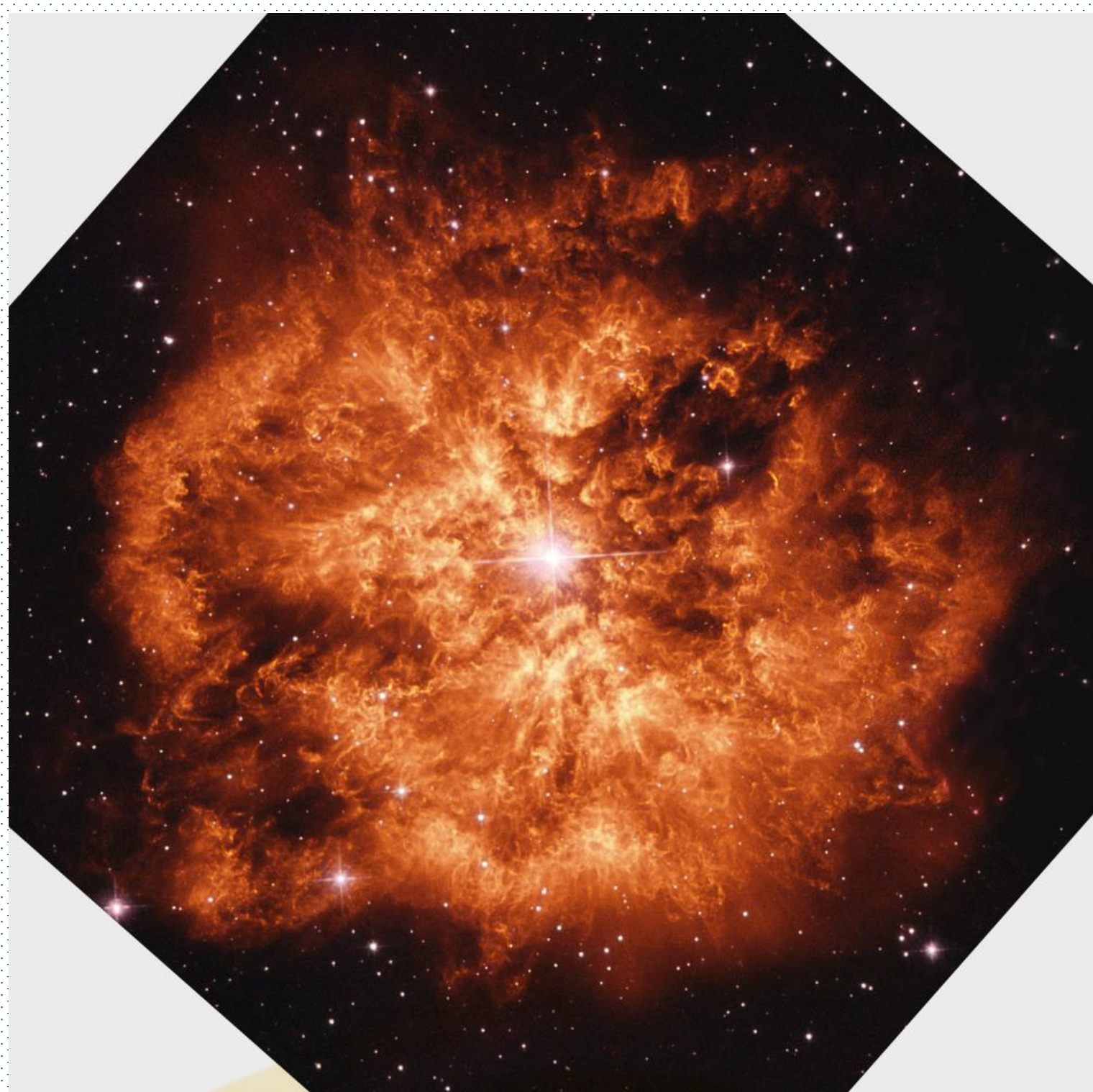
The impact of stellar wind and radiation on neutral knots around a Wolf-Rayet star

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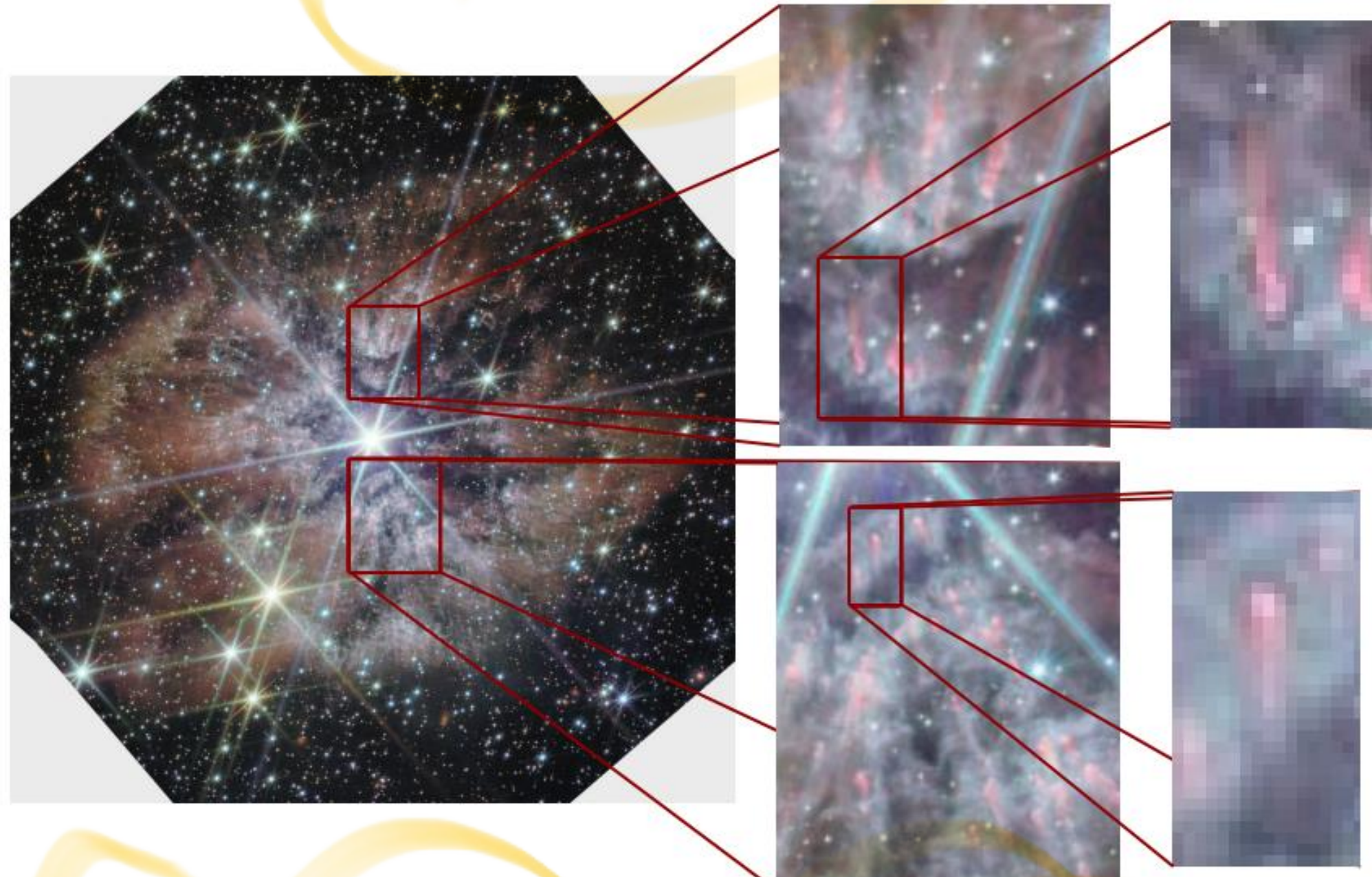
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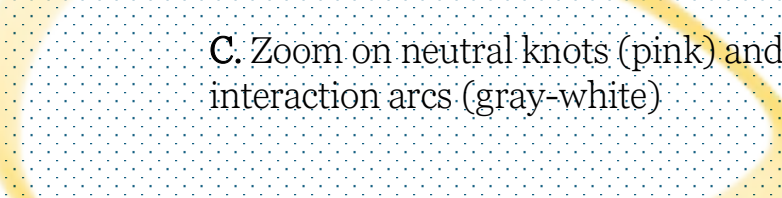
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A. HST WFPC2 H Alpha image of M1-67 nebula (Grodzidier et al 1998)
Image credit: ESA/NASA/Judy Schmidt



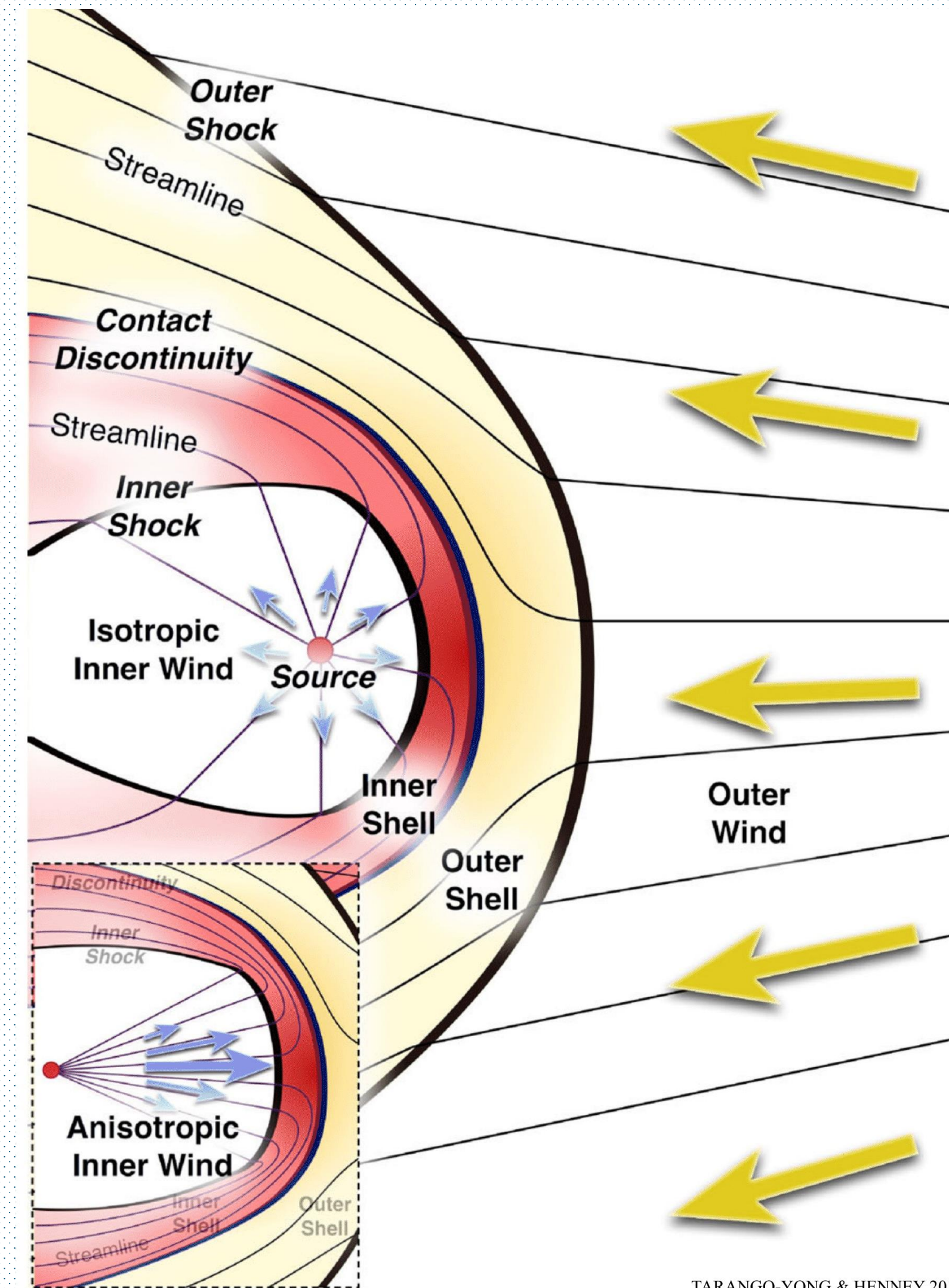
B. JWST NIRCAM image of M1-67, F444W, F335M, F210M, F150W, F090W. Image processing by Judy Schmidt
(http://www.flickr.com/photos/geckzilla/)



C. Zoom on neutral knots (pink) and interaction arcs (gray-white)

Abstract

The circumstellar nebula M1-67 around the runaway WR-124 contains hundreds of small knots, as revealed by recent JWST images. The ionized emission from the nebula shows an intricate pattern of shells and filaments, many of which appear associated with the knot but displaced toward the central star. We proposed a simple model for the nebula in which photoevaporation flows from the irradiated surfaces of the knots interacting with the stellar wind from the Wolf-Rayet star to from hemispherical emission shells. We test this model against JWST images and H alpha HST images of the nebula, confirming that the shells are in pressure balance with both the external stellar wind and the internal photoevaporation flow. We derive a three-dimensional distribution of the knots within the nebula and compare with observations of the knots kinematics, finding a close agreement. The model provides a physical explanation for the observed morphology of the nebula and knots, and suggests that the knots are hydrodynamically shielded from the stellar wind by the photoevaporation flows.



D. Cartoon of the interaction between two supersonic flows. The inner wind is slow and dense, while the outer wind is fast and low density. Only the inner shell is expected to be visible. (Tarango-Yong & Henney 2018)

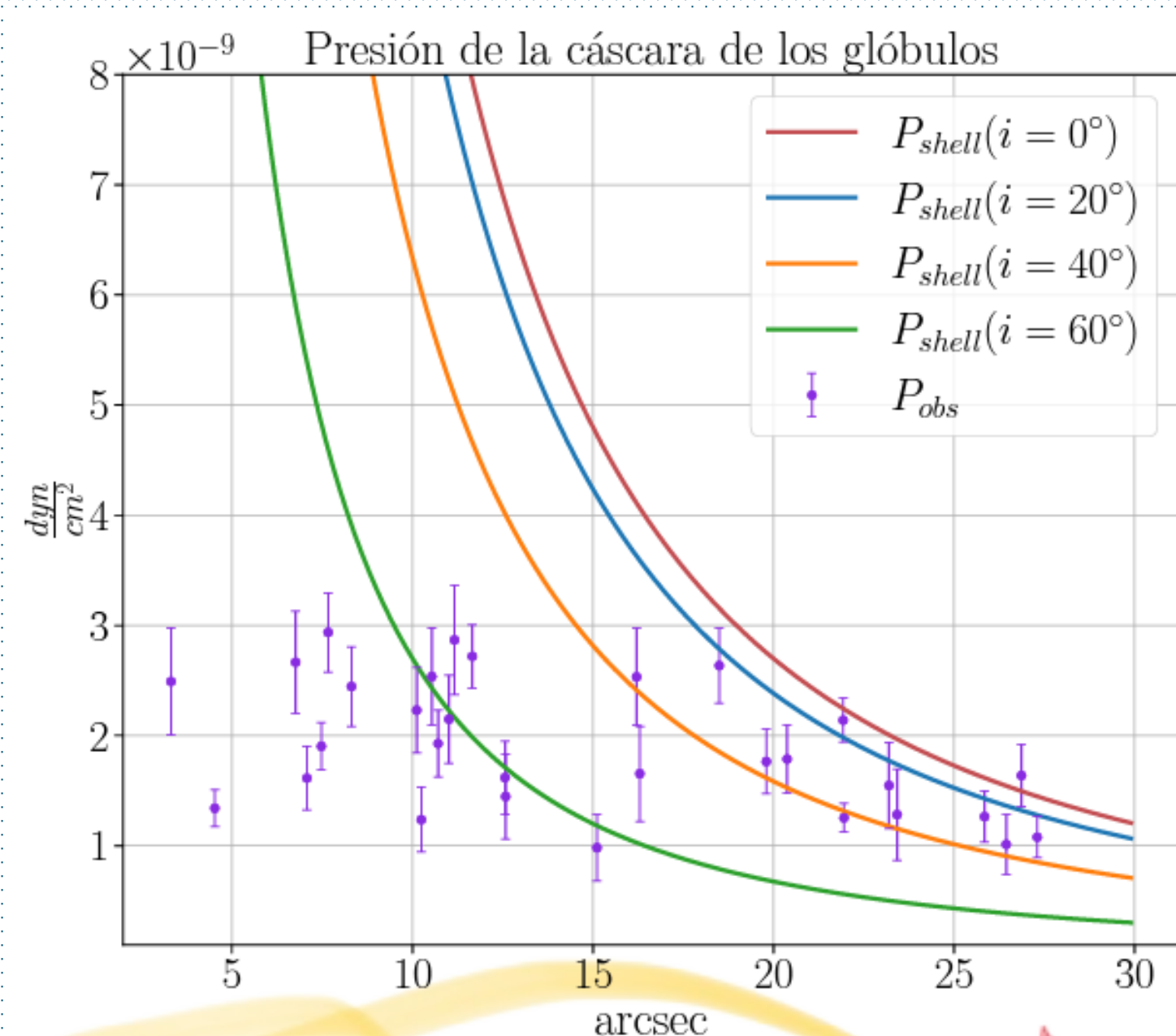
Circumstellar Nebula M1-67

In JWST images we can observe the presence of numerous knots around WR-124, many of which are associated with bright arcs of ionized gas emission, located between the knots and the star. Once we localized these knots and their interaction arcs, we measured the H alpha emission properties of these arcs from archival HST images.

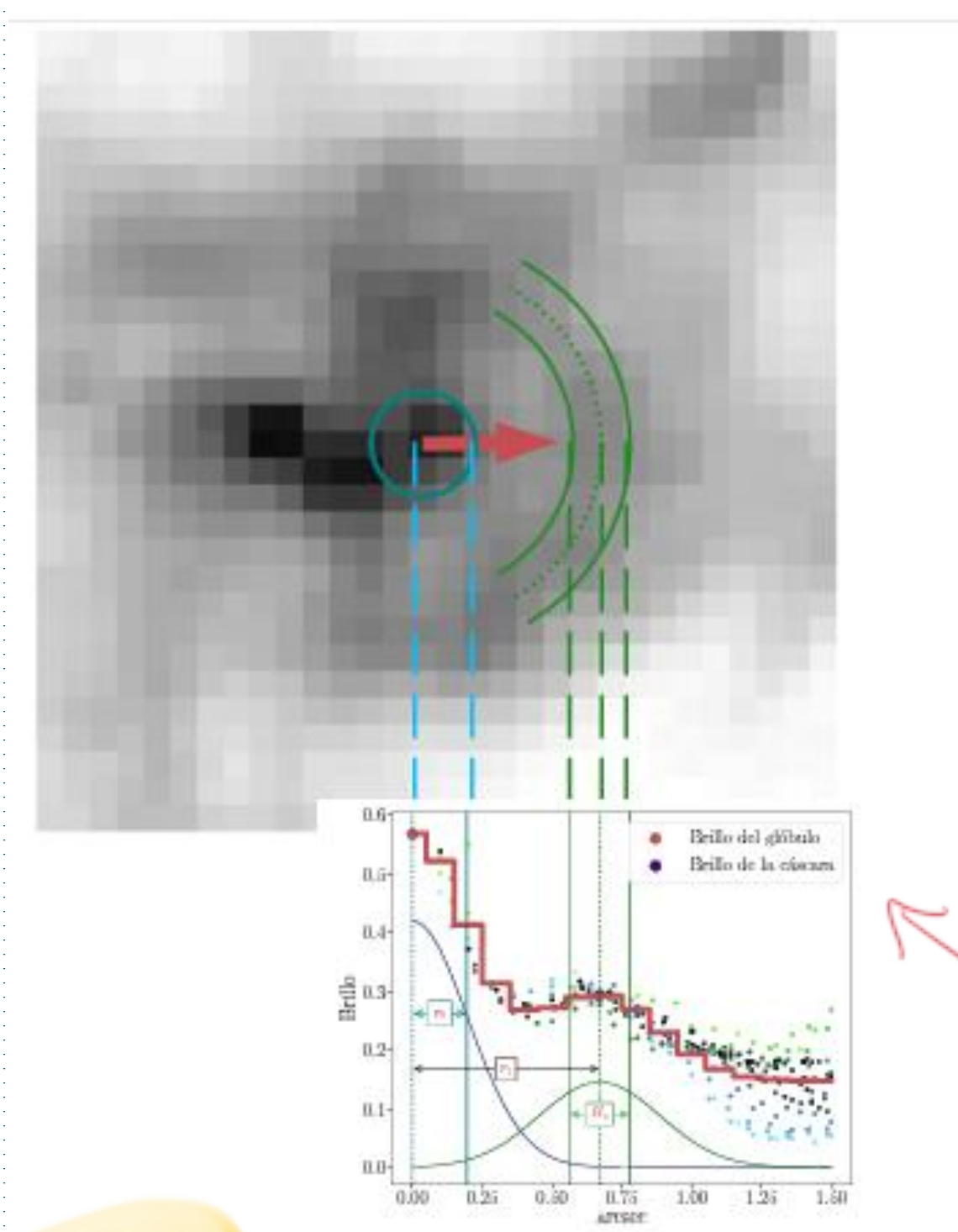
Steady State Hydrodynamic Model

We propose that this emission arc are shock shells resulting from a supersonic wind-wind interaction between the stellar wind from the WR star and a transonic photoevaporation flow from the surfaces of the neutral knots that is induced by the ultraviolet radiation from the star.

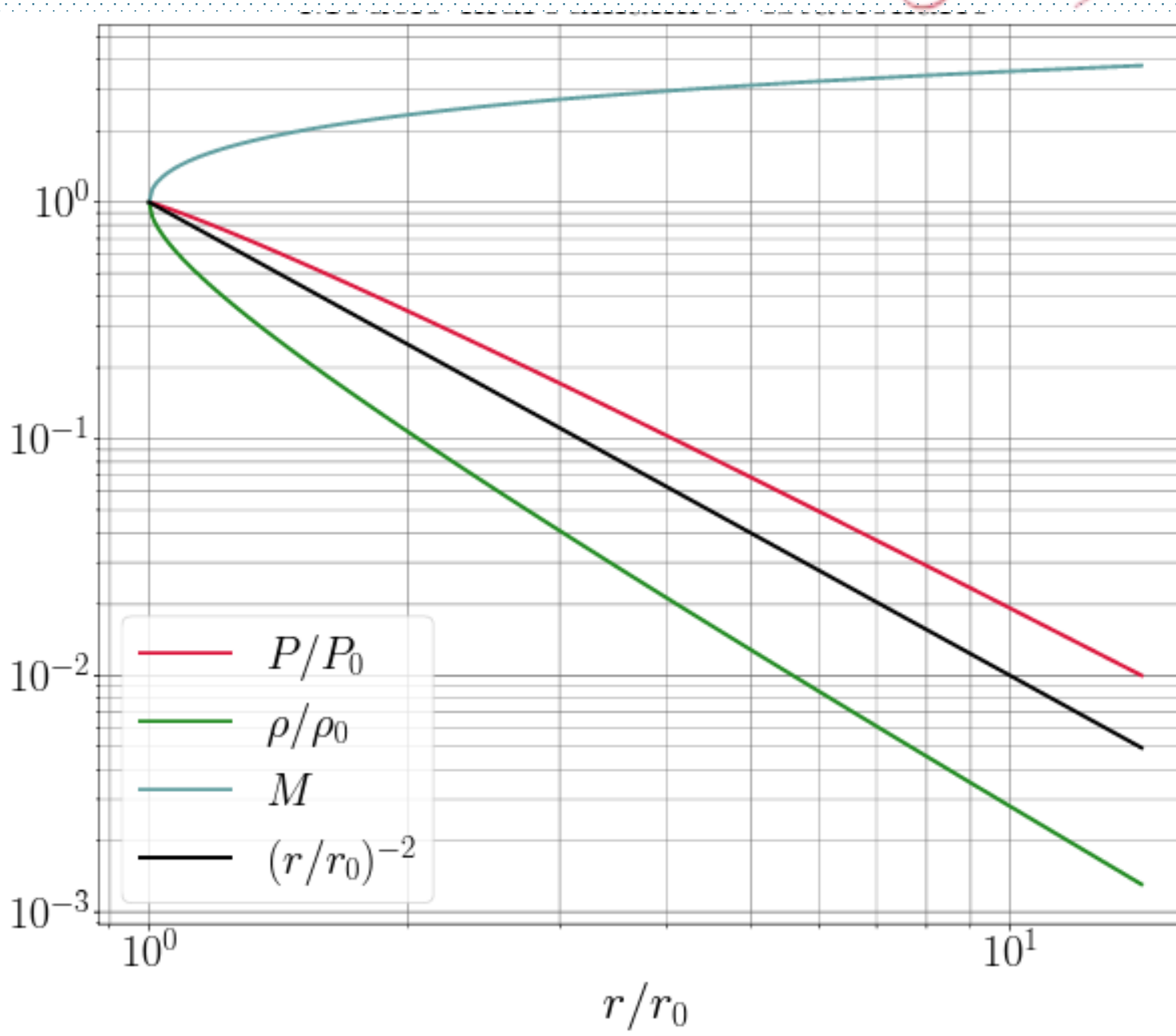
We employ a steady-state hydrodynamic model to calculate the equilibrium point where the RAM pressure of the two winds is in balance. This is also equal to the thermal pressure of the shock shell, which can be observationally estimated from HST images. (Fig F)



E. The pressure of the shell (purple circle) are equal to the RAM pressure (color lines) if we consider an inclination angle.



F. Example of fit of two Gaussians and a constant to brightness profile. With this fit we know the size of the knot, the radius and height of the shell, and the brightness of the knot and the shell.



G. Functions of pressure, density and Mach number normalized as function of r/r_0 .

Comparison between the model and the observations

Once we distinguished the knots in A and B, we made the fit that we can see in F. With the surface brightness we estimated the ionized gas, which is about 1800 cm^{-3} , and compared the shock shells pressures of the knots

$$P_{\text{shell}} = \rho_2 c_s^2$$

With the RAM pressure caused by WR-124

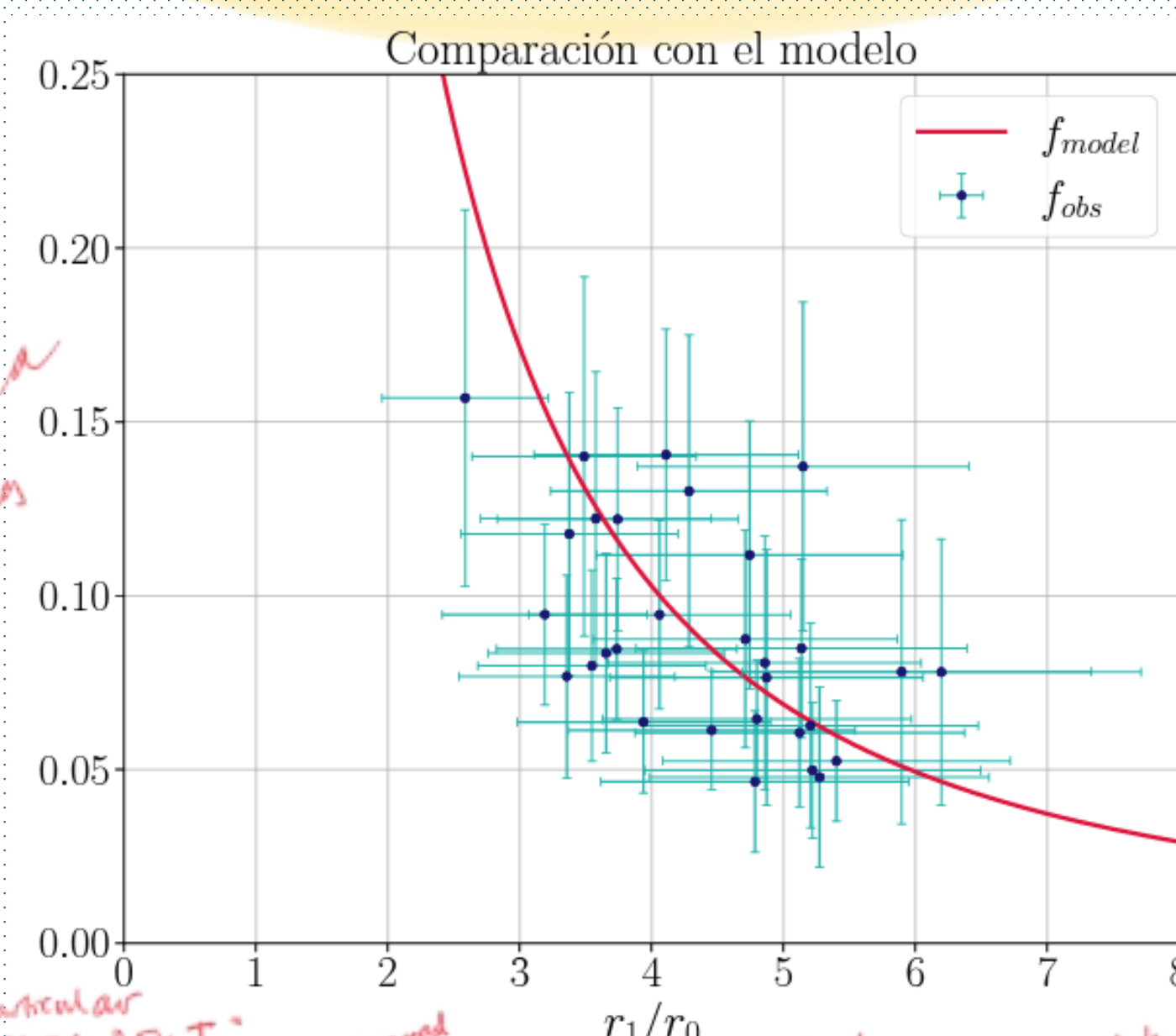
$$P_{\text{RAM}} = \frac{\dot{M} v_{\infty}}{4\pi R^2}$$

Due the shock shells pressures are minor than RAM pressure of the stellar wind, we corrected by an inclination angle (i) to have an equilibrium between the P_{shell} and P_{RAM} , and we can find the real separation between the knots and the star with this inclination angle too.

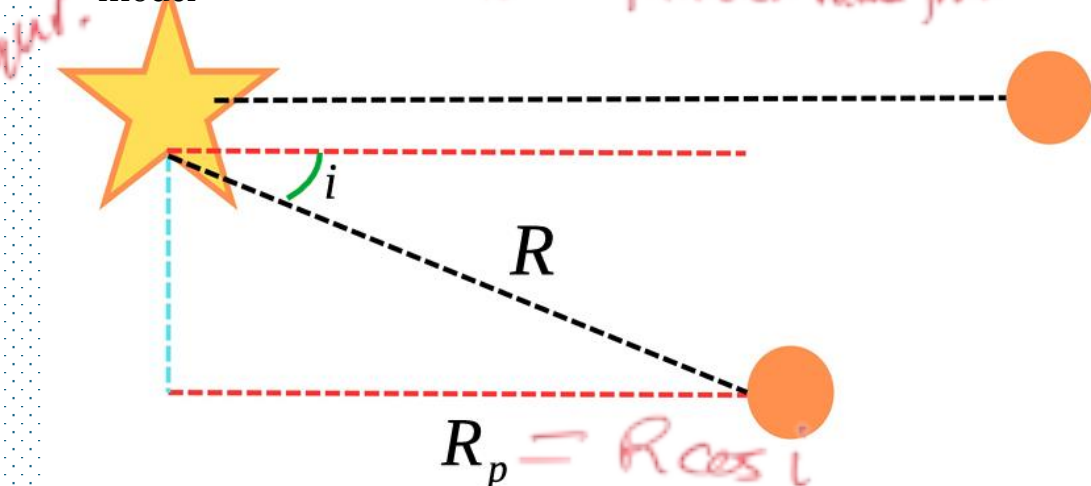
Also, we compared the inner pressure on the surface knots with the shock shells pressures as

$$f_{\text{obs}} = \frac{1}{2} \sqrt{\frac{B_s/B_0}{l_s/l_0}}$$

Where $l = 2\sqrt{2rh}$, B_s and B_0 Are the brightness of the shell and the knot, respectively (F).



H. f_{obs} is the ratio between the photoevaporation flow pressure in the surface of the knots and the pressure of the shells obtained of the observation. f_{model} is the value that we predict with our model.

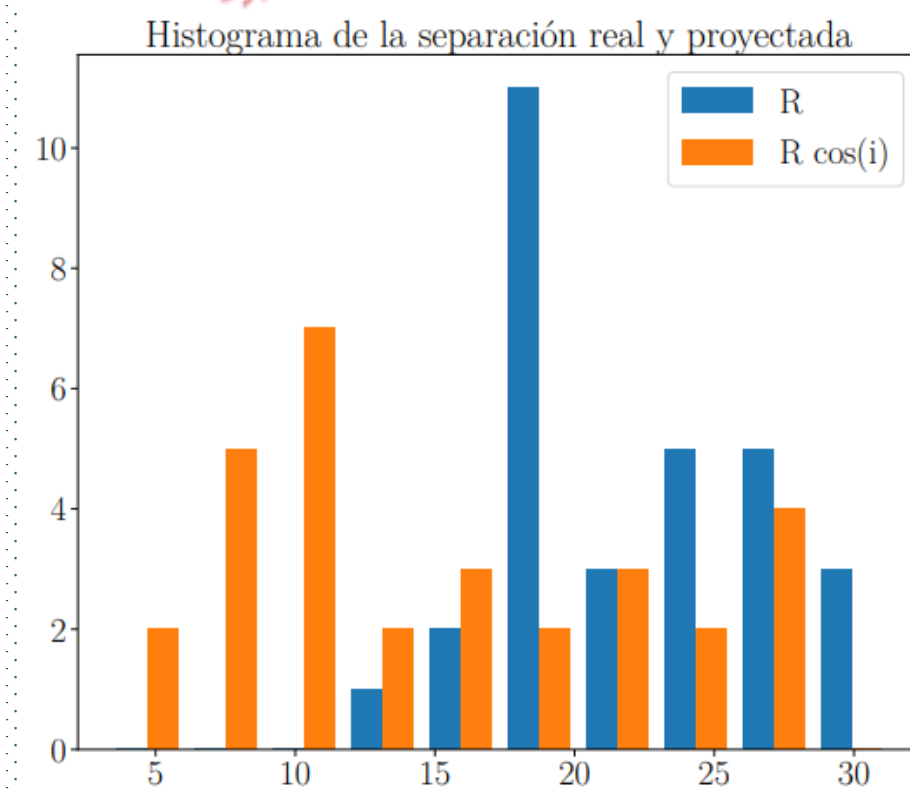


I. Correction by inclination angle i

Conclusions

The measurements are agree with the model that has no free parameters, so we can explain the interaction between the transonic photoevaporation flow and the stellar wind with this simple model.

Additionally, we can know more about the real distribution of the knots around WR-124 that are consisting with another works.



J. Histogram of projected (orange) and real (blue) separation between the star and the knots.

Intenta explicar qué es r_1/r_0 . Escribir las conclusiones en una manera más contundente. No tan vago... Ser más específico en todo si puedes.